Vanishing of first path homology above the random-digraph exponent $-2/3$

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Abstract

Let $G(n,p)$ be the loop-free random digraph in which every ordered edge is present independently with probability p, with reciprocal edges permitted. We prove that its first non-regular path homology over $\mathbb{Q}$ vanishes with probability tending to one if $n^2p^3/\log^4(2n) \to \infty$. Consequently, vanishing holds for $p = n^\alpha$ for every fixed $-2/3 < \alpha \le 0$, as conjectured by Chaplin. The proof uses only boundaries of directed diamonds and transitive triangles. A uniform estimate for the diamond quadratic form shows that its annihilator consists of vertex gradients and the constant edge cochain. A transitive triangle removes the latter direction. The estimates retain the independent ordered-edge model and the non-regular boundary convention throughout.

1 Introduction

Path homology records cycles in a directed graph using relations supplied by allowed directed paths. For random directed graphs it can therefore distinguish directed-network structure from an independent-edge null model. Chaplin [1] studied the first path Betti number in this model and proved its vanishing when $p \gg (\log n/n)^{1/3}$; see Proposition 4.17 of that paper. Conjecture B.1 predicts vanishing throughout the larger power-law range $p = n^\alpha$, $\alpha > -2/3$. The present paper proves this prediction for non-regular path homology over $\mathbb{Q}$ by controlling all edge cochains at once.

We first specify the chain convention. For a digraph G on $[n] = \{1, \dots, n\}$, let $\Lambda_k(\mathbb{Q})$ have as basis all vertex sequences $[v_0, \dots, v_k]$, allowing repetitions, with boundary

$$\partial[v_0, \dots, v_k] = \sum_{j=0}^{k} (-1)^j [v_0, \dots, \widehat{v_j}, \dots, v_k].$$

Let A_k be the subspace spanned by paths whose successive ordered edges belong to G. Put $\Omega_0 = A_0$ and $\Omega_k = A_k \cap \partial^{-1}(A_{k-1})$ for $k \ge 1$. No boundary term with repeated consecutive vertices is discarded. The first non-regular path Betti number over $\mathbb{Q}$ is

$$\beta_1^{\mathrm{nr}}(G) = \dim_{\mathbb{Q}} \frac{\ker(\partial : \Omega_1 \to \Omega_0)}{\partial \Omega_2}.$$

Since G has no loops, $\Omega_1 = A_1$ is the vector space on its ordered edges. The identity $\partial^2 = 0$ gives the indicated quotient.

Write $G(n,p)$ for the digraph with adjacency variables A_{ab}, $a \ne b$, independent Bernoulli(p), and $A_{aa} = 0$. In particular A_{ab} and A_{ba} are independent. All asymptotics below concern deterministic sequences of probabilities as $n \to \infty$.

Theorem 1.1. *Suppose* $0 < p_n \le 1$ *and*

$$\frac{n^2 p_n^3}{\log^4(2n)} \longrightarrow \infty. \tag{1}$$

Then $\mathbb{P}\{\beta_1^{\mathrm{nr}}(G(n,p_n)) = 0\} \longrightarrow 1$.

Corollary 1.2. *For every fixed* $-2/3 < \alpha \le 0$*, a digraph with independent ordered edges of probability* $p_n = n^\alpha$ *satisfies* $\mathbb{P}\{\beta_1^{\mathrm{nr}}(G(n,p_n)) = 0\} \to 1$.

Proof. Here $n^2 p_n^3 = n^{2+3\alpha}$ has a strictly positive exponent and hence dominates $\log^4(2n)$. Apply Theorem 1.1. □

The sufficient condition (1) identifies the conjectured power-law exponent without determining the optimal logarithmic factor. Neither the critical exponent $\alpha = -2/3$ nor convergence uniform in α is asserted. The conclusion concerns rational homology; it does not assert the absence of integral torsion.

The mechanism is visible from two elementary boundaries. A directed diamond has two length-two paths with the same endpoints; subtracting them cancels the possibly missing shortcut edge. Its boundary equates the two path sums. Vertex gradients satisfy every such relation, as does the cochain that is one on each edge. We show that these are the only solutions with high probability. A single transitive triangle then excludes the constant cochain. The main difficulty is uniformity: a cochain under examination can depend on the entire graph. Operator estimates, rather than estimates for a fixed cochain, provide the required simultaneous conclusion.

2 The diamond quadratic form

In this section G is a fixed loop-free digraph, with edge set $E = \{(a,b) : A_{ab} = 1\}$. We work first over $\mathbb{R}$. For an edge cochain $f \in \mathbb{R}^E$ and a vertex function $h \in \mathbb{R}^n$, define $(\delta h)_{ab} = h_b - h_a$. The space left undetected by the diamond relations will be

$$W = \operatorname{im}\delta + \mathbb{R}\mathbf{1}_E. \tag{2}$$

All vector norms and inner products are Euclidean, and all matrix norms are their induced operator norms. The letter J denotes an all-one matrix of the dimensions specified by its indices.

A diamond consists of four distinct vertices a, b, c, e and edges $a \to b$, $b \to e$, $a \to c$, $c \to e$. It is counted once for ordered endpoints (a, e) and an unordered pair of middle vertices $\{b, c\}$. Its boundary row and energy are

$$q_{a;b,c;e} = [a,b] + [b,e] - [a,c] - [c,e], \qquad \mathcal{E}_G(f) = \sum_{\text{diamonds}} \langle q_{a;b,c;e}, f\rangle^2.$$

The sign chosen for the unordered middle pair is immaterial. The row is the literal boundary of $[a,b,e] - [a,c,e]$: the two occurrences of $[a,e]$ cancel. Thus this two-chain belongs to the non-regular Ω_2, whether or not $a \to e$ is an edge. Every row annihilates W by telescoping and by $1+1-1-1 = 0$.

To expand the energy, put

$$O_v = \{b : A_{vb} = 1\}, \quad I_v = \{a : A_{av} = 1\}, \quad x_v = (f_{vb})_{b \in O_v}, \quad y_v = (f_{av})_{a \in I_v}.$$

Their sums will be denoted by $r_v = \sum_{b \in O_v} f_{vb}$, $c_v = \sum_{a \in I_v} f_{av}$, and $s = \sum_{(a,b) \in E} f_{ab}$. Empty vectors and sums are interpreted as zero. Let D_{ab} be the number of diamonds containing an edge (a, b). The two fork matrices have zero diagonal and, off the diagonal, entries

$$(P_a)_{b,c} = \sum_{e \notin \{a,b,c\}} A_{be} A_{ce} \qquad (b, c \in O_a,\ b \neq c), \tag{3}$$

$$(Q_e)_{b,c} = \sum_{a \notin \{e,b,c\}} A_{ab} A_{ac} \qquad (b, c \in I_e,\ b \neq c). \tag{4}$$

For an adjacent pair of edges through v, use the rectangular matrix

$$(C_v)_{a,b} = \mathbf{1}_{a \neq b} \sum_{c \notin \{a,b,v\}} A_{ac} A_{cb} \quad (a \in I_v,\ b \in O_v). \tag{5}$$

Finally the opposite-edge matrix, indexed by actual edges, is

$$H_{(a,b),(c,e)} = \mathbf{1}_{\{a,b,c,e\} \text{ has size } 4}\, A_{ac} A_{be} \quad ((a, b), (c, e) \in E). \tag{6}$$

This matrix need not be symmetric; its operator norm will suffice.

Lemma 2.1. *For every real edge cochain,*

$$\mathcal{E}_G(f) = \sum_E D_{ab} f_{ab}^2 - \sum_v x_v^T P_v x_v - \sum_v y_v^T Q_v y_v + 2 \sum_v y_v^T C_v x_v - 2 f^T H f. \tag{7}$$

Proof. For one diamond write its four values as $x = f_{ab}$, $y = f_{be}$, $u = f_{ac}$, $z = f_{ce}$. Expansion of $(x + y - u - z)^2$ gives diagonal squares, positive adjacent products $2xy + 2uz$, negative source and sink products $-2xu - 2yz$, and negative opposite products $-2xz - 2yu$. The diagonal multiplicity is D_{ab}. Each fork matrix counts both orders of its two middle vertices, producing its coefficient -2. The matrix C_v counts each adjacent product once. The two orders of (b, c) in H produce the two opposite products. These are precisely the five terms in (7). □

We will prove that (7) is close, uniformly in f, to a quadratic form depending only on $\|f\|$, $r - c$ and s. The needed estimates account for the dependence between a selected edge and its neighboring diamond relations.

3 Concentration ingredients

Fix a sequence satisfying (1), and abbreviate

$$\ell = \log(2n), \qquad d = n^2 p^3, \qquad \mu = (n-3)p^2, \qquad \varepsilon_n = \sqrt{d}\,\ell^2 + d\sqrt{\frac{\ell}{np}} + np^2\ell^2 + \ell^3. \tag{8}$$

In particular $d \to \infty$ and $np = (nd)^{1/3}$ grows faster than every fixed power of ℓ. Constants denoted C_L depend only on a fixed positive number L and may increase finitely many times. A failure bound $O_L(n^{-L})$ has this same uniformity in n, p and the fixed vertex sets under consideration.

The only external concentration input is matrix Bernstein in the following two forms [2, Theorems 1.4 and 1.6 of the author preprint]. For independent mean-zero self-adjoint $q \times q$ matrices X_i with $\|X_i\| \le R$, put $\sigma^2 = \left\|\sum_i \mathbb{E} X_i^2\right\|$. Then

$$\mathbb{P}\left\{\left\|\sum_i X_i\right\| \ge t\right\} \le 2q \exp\left\{-\frac{t^2}{2(\sigma^2 + Rt/3)}\right\} \qquad (t \ge 0). \tag{9}$$

The two-sided form follows by applying the source's upper-eigenvalue bound to both X_i and $-X_i$. For independent mean-zero $q_1 \times q_2$ matrices Z_i with $\|Z_i\| \le R$, the norm bound has prefactor $q_1 + q_2$ and the same exponent, with

$$\sigma^2 = \max\left\{ \left\| \sum_i \mathbb{E} Z_i Z_i^T \right\|, \left\| \sum_i \mathbb{E} Z_i^T Z_i \right\| \right\}.$$

Taking dimension one gives, for a sum of independent Bernoulli variables with mean M, deviation at most $C_L(\sqrt{M\ell} + \ell)$ with failure $O_L(n^{-L})$. Throughout the proof dimensions are at most n^2, so the logarithmic factor absorbs the dimension prefactors and polynomial union bounds by choosing L larger at the outset.

3.1 Independent row and column restrictions

The next lemma gains a factor p when both coordinate sets are sampled. It will be used only where the sampled edge families are independent of the matrix being restricted.

Lemma 3.1. *Let M be a deterministic real matrix with both dimensions at most n^2. Let R and S be independent diagonal matrices whose diagonal entries are independent Bernoulli(p) variables. If*

$$a = \|M\|, \quad u = \max_i \|M_{i,*}\|_2, \quad v = \max_j \|M_{*,j}\|_2, \quad b = \max_{i,j} |M_{ij}|,$$

then, with failure $O_L(n^{-L})$,

$$\|RMS\| \le pa + C_L\sqrt{p\ell}(u + v) + C_L \ell b. \tag{10}$$

The same conclusion holds conditionally for a random M independent of R and S.

Proof. First consider row selection of a fixed matrix B with rows b_i. Writing $u_B = \max_i \|b_i\|_2$, the centered Gram matrix is

$$B^T R B - pB^T B = \sum_i (\xi_i - p) b_i^T b_i.$$

Its independent summands have norm at most u_B^2. Since $(b_i^T b_i)^2 = \|b_i\|_2^2 \, b_i^T b_i$, their variance matrix has norm at most $p u_B^2 \|B\|^2$. The upper-eigenvalue bound in (9) gives

$$\|RB\|^2 \le p\|B\|^2 + C_L u_B \sqrt{p\ell}\,\|B\| + C_L \ell u_B^2.$$

Taking square roots and enlarging C_L yields

$$\|RB\| \le \sqrt{p}\,\|B\| + C_L\sqrt{\ell}\, u_B. \tag{11}$$

Apply (11) to M^T to obtain $\|MS\| \le \sqrt{p}\, a + C_L\sqrt{\ell}\, v$. For a fixed row, $\|(MS)_{i,*}\|_2^2 = \sum_j \zeta_j |M_{ij}|^2$. The centered summands are bounded by b^2 and have total variance at most pb^2u^2. Scalar Bernstein and a union bound over the rows give

$$\max_i \|(MS)_{i,*}\|_2 \le \sqrt{p}\, u + C_L\sqrt{\ell}\, b.$$

Conditional on S, apply (11) to MS using independent R. Substituting the preceding two inequalities proves (10). Conditioning on an independent random M proves the last assertion. □

3.2 Products on disjoint vertex blocks

For disjoint vertex sets V_i, write $B_{ij} = (A_{ab})_{a\in V_i, b\in V_j}$. The entries of each block are independent Bernoulli(p); blocks with different ordered source–target sets are independent.

Lemma 3.2. *Fix three disjoint vertex sets of sizes at most n, and put $M = B_{12}B_{23} - |V_2|p^2J$. For independent Bernoulli(p) coordinate selectors R, S, also independent of the blocks,*

$$\|RMS\| \le C_L\varepsilon_n \tag{12}$$

with failure $O_L(n^{-L})$. For four disjoint vertex sets the same bound holds with $M = B_{13}\otimes B_{24} - p^2J$, where the row coordinates are pairs in $V_1\times V_2$ and the column coordinates are pairs in $V_3\times V_4$.

Proof. For any one of these blocks let $X = B - pJ$. Entrywise rectangular Bernstein has summand norm at most 1 and variance at most np, giving

$$\|X\| \le C_L\sqrt{np\ell} \tag{13}$$

because $np \gg \ell$. Scalar Bernstein and union bounds give degrees at most C_Lnp and absolute centered row and column sums at most $C_L\sqrt{np\ell}$. Each row and column of B and X has norm at most $C_L\sqrt{np}$: for X, use $\sum_j(B_{ij}-p)^2 \le 2\sum_j B_{ij} + 2np^2$. These bounds are uniform in every fixed choice of the vertex sets, including empty sets.

For three sets, the centered product is

$$M = pJX_{23} + pX_{12}J + X_{12}X_{23}.$$

Using (13) and $\|J\| \le n$ gives

$$\|M\| \le C_L\big(np\ell + n^{3/2}p^{3/2}\sqrt{\ell}\big). \tag{14}$$

For each row the three terms are bounded respectively by $p\sqrt{n}\,\|X_{23}\|$, $p\sqrt{n}$ times the magnitude of a row sum of X_{12}, and a row norm of X_{12} times $\|X_{23}\|$. Columns satisfy the reversed estimates. Thus

$$\max\{\text{row norms, column norms of } M\} \le C_Lnp\sqrt{\ell}. \tag{15}$$

Each entry of $B_{12}B_{23}$ is Binomial($|V_2|, p^2$): different middle vertices use distinct edges of the two disjoint blocks. Consequently

$$\max|M_{ij}| \le C_L\big(\sqrt{np^2\ell} + \ell\big) \le C_L(np^2 + \ell). \tag{16}$$

Lemma 3.1 applied to (14)–(16) now yields

$$\|RMS\| \le C_L\left(np^2\ell + n^{3/2}p^{5/2}\sqrt{\ell} + \sqrt{d}\,\ell + \ell^2\right) \le C_L\varepsilon_n.$$

Here $n^{3/2}p^{5/2}\sqrt{\ell} = d\sqrt{\ell/(np)}$.

For the tensor product expand instead

$$M = X_{13}\otimes X_{24} + pJ\otimes X_{24} + pX_{13}\otimes J.$$

The inequality $\|U\otimes V\| \le \|U\|\,\|V\|$ gives (14) again. The uncentered tensor has row and column norms bounded by products of two block row or column norms, at most C_Lnp. A row or column of the subtracted matrix p^2J has norm at most $np^2 \le np$. Entry magnitudes of M are at most 2. Lemma 3.1 therefore gives

$$\|RMS\| \le C_L\left(np^2\ell + n^{3/2}p^{5/2}\sqrt{\ell} + \sqrt{d\ell} + \ell\right) \le C_L\varepsilon_n.$$

The tensor norm inequality follows, for example, by reshaping a vector as a matrix and using $\|UZV\|_F \le \|U\|\,\|Z\|_F\,\|V\|$. □

3.3 Averaging the cuts

We use random vertex labels to place the selected edges and connector edges in independent blocks. The following finite averaging observation recovers the original matrix without demanding that every partition be good.

Lemma 3.3. *Let k, t be fixed positive integers. Suppose a graph-dependent matrix T and matrices T_χ associated with independent uniform k-labels on vertices satisfy $T = k^t \mathbb{E}_\chi T_\chi$. The matrices are embedded in a common coordinate space for each fixed graph. Suppose every $\|T_\chi\| \le 2n^2$, and for each fixed labeling*

$$\mathbb{P}_G\{\|T_\chi\| > C_L \varepsilon_n\} \le O_L(n^{-L}).$$

Then, except with probability $O_L(n^{-L/2})$,

$$\|T\| \le k^t\big(C_L \varepsilon_n + 2n^{2-L/2}\big).$$

Proof. Let $h(G)$ be the fraction of labelings with $\|T_\chi\| > C_L\varepsilon_n$. Finite-sum interchange gives $\mathbb{E}_G h(G) \le O_L(n^{-L})$. By Markov's inequality, $h(G) \le n^{-L/2}$ outside an event of probability $O_L(n^{-L/2})$. On that complement the triangle inequality for the finite average gives the stated bound. □

4 Uniform diamond estimates

We now return to the matrices (3)–(6) indexed by the random neighborhoods or edge set of $G(n,p)$.

Proposition 4.1. *Under (1), with probability tending to one, the following bounds hold simultaneously:*

$$\max_{(a,b)\in E} |D_{ab} - 2d| \le C\varepsilon_n, \tag{17}$$

$$\max_a \|P_a - \mu(J-I)\| + \max_e \|Q_e - \mu(J-I)\| \le C\varepsilon_n, \tag{18}$$

$$\max_v \|C_v - \mu J\| \le C\varepsilon_n, \tag{19}$$

$$\big\|H - p^2 J_{E,E}\big\| \le C\varepsilon_n. \tag{20}$$

Here C is an absolute constant.

We prove the mixed and opposite bounds first, then the fork and diagonal bounds. Take, for example, $L = 100$ in the concentration ingredients; enlarging initial tail exponents accommodates all finite union bounds.

4.1 Mixed terms and opposite edges

Fix a vertex v and assign the other vertices independent uniform labels from $\{1,2,3\}$. Let V_i be their label classes. Retain in C_v only triples (a,c,b) of labels $(1,2,3)$, and subtract $|V_2|p^2$ in the retained endpoint rectangle. This centered cut is the restriction

$$R_{I_v}\big(B_{12}B_{23} - |V_2|p^2 J\big)R_{O_v}.$$

The row selectors are A_{av} for $a \in V_1$, the column selectors are A_{vb} for $b \in V_3$. These two families are mutually independent and independent of B_{12}, B_{23}, since the latter blocks avoid v. Lemma 3.2 applies for each fixed labeling.

For a fixed graph, pad this cut with zero rows and columns into the coordinate space $I_v \times O_v$ and average. Each distinct triple (a, c, b) has the prescribed labels with probability $1/27$. Consequently 27 times the average centered cut is exactly

$$C_v - \mu(J - F_v),$$

where $(F_v)_{a,b} = \mathbf{1}_{a=b}$ on $I_v \times O_v$. Indeed, for $a \neq b$ there are $n-3$ eligible middle vertices; for $a = b$ the label requirements are impossible. The deterministic cut norm is at most $2n^2$, because its dimensions are at most n and its entry magnitudes at most $2n$. Lemma 3.3 and a union bound over v yield $\max_v \|C_v - \mu(J - F_v)\| \leq C\varepsilon_n$. Since $\|F_v\| \leq 1$ and $\mu \leq np^2 \leq \varepsilon_n$, this proves (19).

For H use four labels, assigning (a, b, c, e) the labels $(1, 2, 3, 4)$. The connector matrix is $B_{13} \otimes B_{24}$, indexed by pairs (a, b) and (c, e). Restriction to actual edge pairs uses independent selectors A_{ab} in block 12 and A_{ce} in block 34. They are mutually independent and independent of connector blocks 13 and 24. Thus Lemma 3.2 applies to the centered cut $B_{13} \otimes B_{24} - p^2 J$ with these pair-coordinate selectors. After zero-padding in the edge coordinate space, each four-distinct tuple occurs with probability $1/256$ and the cut norm is at most $2n^2$. Lemma 3.3 gives

$$\left\| H - p^2 J^{(4)}_{E,E} \right\| \leq C\varepsilon_n,$$

where $J^{(4)}$ is one for edge pairs with four distinct vertices and zero otherwise.

With probability $1 - O_L(n^{-L})$, all in- and out-degrees are at most $C_L np$, by scalar Bernstein. On this event each actual edge shares a vertex with at most $C_L np$ other actual edges. Hence the symmetric nonnegative matrix $J_{E,E} - J^{(4)}_{E,E}$ has maximum row sum at most $C_L np$, and its norm has the same upper bound. Multiplication by p^2 costs at most $C_L np^3 \leq C_L \varepsilon_n$, proving (20). For completeness, a matrix with entry magnitudes having row sums at most R and column sums at most S has norm at most $\sqrt{RS}$, by weighted Cauchy–Schwarz. This justifies the row-sum bound without a symmetry assumption on H itself.

4.2 Forks

Fix a and condition only on its outgoing neighborhood $O_a = S$. On the degree event, $m = |S| \leq C_L np$. For $e \neq a$ let $u_e = (A_{be})_{b \in S}$ and $Z_e = u_e u_e^T - \mathrm{diag}(u_e)$. These columns are independent: their edges have distinct ordered target vertices and none has source a. When $e \in S$ its self-loop coordinate is deterministically zero; the remaining coordinates are independent Bernoulli(p). Reciprocal edges in different columns remain independent. We have

$$P_a = \sum_{e \neq a} Z_e, \qquad \mathbb{E}(P_a \mid S) = \mu(J - I).$$

For each off-diagonal pair (b, c) the $n-3$ permissible witnesses exclude a, b, c, which explains the exact mean.

For one such column, the diagonal entries of $\mathbb{E}Z_e^2$ are at most mp^2, and the off-diagonal entries are at most mp^3. This follows from $(Z_e)_{ij} = (u_e)_i (u_e)_j$ for $i \neq j$: a diagonal square uses two distinct Bernoulli coordinates, while an off-diagonal square uses three. The entries are nonnegative, so their maximum row sum is at most $mp^2 + m^2 p^3$.

Truncate each Z_e separately to zero if the column support satisfies $|\operatorname{supp} u_e| > C_L(mp + \ell)$, and call the result $\widetilde{Z}_e$. Scalar Bernstein bounds the probability of each truncation by $O_L(n^{-L})$. Truncation preserves column independence, and the total mean lost has norm at most $O_L(nm\, n^{-L})$,

since $\|Z_e\| \le m$. Every entry of $\mathbb{E}\widetilde{Z}_e^2$ is bounded by the corresponding nonnegative entry of $\mathbb{E}Z_e^2$. Thus its norm is at most $mp^2 + m^2p^3$, by the same row-sum bound. In addition,

$$\mathbb{E}(\widetilde{Z}_e - \mathbb{E}\widetilde{Z}_e)^2 = \mathbb{E}\widetilde{Z}_e^2 - (\mathbb{E}\widetilde{Z}_e)^2 \preccurlyeq \mathbb{E}\widetilde{Z}_e^2.$$

The centered variance sum therefore has norm at most

$$n(mp^2 + m^2p^3) \le C_L d(1 + np^2).$$

Each centered summand has norm at most $C_L(np^2 + \ell)$, because the retained support count bounds $\left\|\widetilde{Z}_e\right\|$ and $mp \le C_L np^2$. Applying (9) gives

$$\|P_a - \mu(J - I)\| \le C_L \left(\sqrt{d(1 + np^2)\ell} + (np^2 + \ell)\ell\right) + o(1) \le C_L \varepsilon_n. \tag{21}$$

Here $\sqrt{d\, np^2 \ell} = d\sqrt{\ell/(np)}$. The truncated and untruncated sums differ only on the union of the column truncation events. The conditional estimates are uniform in every S of the permitted size, so they may be averaged over S and union bounded over a. Empty or singleton neighborhoods cause no difficulty. Reversing all arrows gives the corresponding bound for Q_e. This proves (18).

4.3 Diagonal multiplicities

Fix (a, b) and condition on $A_{ab} = 1$. The diamonds in which this edge is on the first step are counted by

$$\sum_{\substack{c,e \notin \{a,b\} \\ c \ne e}} A_{ac} A_{be} A_{ce}.$$

Expose just the outgoing rows of a, b, and put $S = O_a \setminus \{b\}$ and $T = O_b \setminus \{a\}$. The remaining edges $c \to e$ with $(c, e) \in S \times T$, $c \ne e$, have source different from a, b. They are unexposed independent Bernoulli(p) variables. The conditional count is consequently

$$\text{Binomial}\big(|S||T| - |S \cap T|, p\big).$$

Under $A_{ab} = 1$, scalar Bernstein for the other row entries gives

$$|S|, |T| = np + O_L(\sqrt{np\ell} + \ell + 1)$$

outside an event of probability $O_L(n^{-L})$. Since $|S \cap T| \le \min(|S|, |T|)$ and $np \gg \ell$, the conditional mean differs from d by at most $C_L d\sqrt{\ell/(np)}$. Another conditional scalar Bernstein bound contributes at most $C_L(\sqrt{d\ell} + \ell)$. The first-step count is therefore $d + O_L(\varepsilon_n)$.

For diamonds in which (a, b) is on the second step, expose instead the incoming edge families of a, b. The remaining connectors have target different from a, b, so the same conditional calculation gives $d + O_L(\varepsilon_n)$. These two types are disjoint and their sum is D_{ab}; they need not be independent. The failure bound is uniform conditional on every possible present edge, and a union bound over at most n^2 edges proves (17). No conditioning on a global degree event is added to the conditional binomial law. This finishes the proof of Proposition 4.1.

4.4 The uniform quadratic-form estimate

The estimates just proved hold on a single graph event and therefore apply simultaneously to graph-dependent cochains.

Proposition 4.2. *With probability tending to one under (1), every $f \in \mathbb{R}^E$ satisfies*

$$\left|\mathcal{E}_G(f) - \left((2d+2\mu)\,\|f\|^2 - \mu\,\|r-c\|^2 - 2p^2s^2\right)\right| \le C\varepsilon_n\,\|f\|^2\,. \tag{22}$$

In particular, on that event and for all sufficiently large n,

$$f \perp W \quad\Longrightarrow\quad \mathcal{E}_G(f) \ge d\,\|f\|^2\,. \tag{23}$$

Proof. In (7), replacing the five matrix families by their main terms in Proposition 4.1 gives respectively

$$2d\,\|f\|^2\,, \quad -\mu(\|r\|^2 - \|f\|^2), \quad -\mu(\|c\|^2 - \|f\|^2), \quad 2\mu\langle c, r\rangle, \quad -2p^2s^2.$$

Their sum is the main form in (22). For each fork family, the error is at most $C\varepsilon_n\,\|f\|^2$ because every edge appears in exactly one outgoing vector and exactly one incoming vector. For the mixed term use

$$\sum_v \|x_v\|\,\|y_v\| \le \left(\sum_v \|x_v\|^2\right)^{1/2}\left(\sum_v \|y_v\|^2\right)^{1/2} = \|f\|^2\,.$$

The diagonal and opposite-edge errors have the same bound by (17) and (20). This proves (22) uniformly in f.

For $f \perp W$, pairing with gradients gives $\langle f, \delta h\rangle = \sum_v h_v(c_v - r_v) = 0$ for every h, so $r = c$. Pairing with $\mathbf{1}_E$ gives $s = 0$. Finally

$$\frac{\varepsilon_n}{d} = \frac{\ell^2}{\sqrt{d}} + \sqrt{\frac{\ell}{np}} + \frac{\ell^2}{np} + \frac{\ell^3}{d} \longrightarrow 0$$

by (1) and $np = (nd)^{1/3}$. Thus $(2d+2\mu) - C\varepsilon_n \ge d$ for all sufficiently large n, which proves (23). □

5 Rational homology vanishes

We complete the proof of Theorem 1.1. Let K be the space of real edge cochains annihilating every diamond row. We already have $W \subseteq K$. On the event of Proposition 4.2, any $f \in K \setminus W$ would have a nonzero orthogonal component $f_0 \in W^\perp$. Since subtracting a member of W preserves all diamond evaluations, $f_0 \in K$ and $\mathcal{E}_G(f_0) = 0$. This contradicts (23). Consequently

$$K = \operatorname{im}\delta + \mathbb{R}\mathbf{1}_E \tag{24}$$

with probability tending to one.

The graph also contains a transitive triangle with probability tending to one. To see this, fix $v = 1$. Scalar Bernstein gives $|O_v| = (1 + o(1))np$ with high probability. Conditional on O_v, all ordered edges between distinct vertices of O_v are independent Bernoulli(p): none has source v. Their probability of all being absent is at most

$$(1-p)^{|O_v|(|O_v|-1)} \le \exp\{-p|O_v|(|O_v|-1)\} = \exp\{-(1+o(1))d\} \longrightarrow 0.$$

An edge $a \to b$ inside O_v supplies $v \to a$, $a \to b$, $v \to b$. The allowed two-path $[v, a, b]$ has allowed boundary

$$[a, b] - [v, b] + [v, a].$$

Every gradient evaluates to zero on this row, whereas $\mathbf{1}_E$ evaluates to 1. Combining this triangle with (24), the common annihilator of the diamond and transitive-triangle rows is exactly $\operatorname{im} \delta$.

Let $\mathcal{B} \subseteq \mathbb{R}^E$ be the span of these rows and let $\partial_1 : \mathbb{R}^E \to \mathbb{R}^n$ be the vertex-incidence boundary. All the rows are cycles, and

$$\mathcal{B}^\perp = \operatorname{im} \delta = (\ker \partial_1)^\perp.$$

Finite-dimensional duality yields $\mathcal{B} = \ker \partial_1$. The incidence matrix and the matrix of the displayed boundary rows have integer entries, so each has the same rank over $\mathbb{Q}$ and $\mathbb{R}$: rank is the largest order of a nonzero minor. Therefore the rational span of these valid non-regular boundaries is the full rational edge-cycle space. It is contained in $\partial\Omega_2$, which is itself contained in $\ker \partial_1$. Their equality proves $\beta_1^{\mathrm{nr}}(G(n, p_n)) = 0$ on an event of probability tending to one, as required.

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References

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